

The *Diversity–Strength* Frontier

What actually separates a validated 0.95390 from the leaderboard's top scores? An honest, nine-model investigation into why an ensemble stops improving — and why the answer is a wall, not a trick.

- Binary classification · ROC-AUC
- 439,140 rows · 80/20 imbalance
- Top 15% finish

01 – THE QUESTION

A gap that wouldn't close

My best honest submission scored **0.95390**. The top of the leaderboard sat at **0.95493** — about ten basis points higher. Ten thousandths of a point. On a synthetic tabular problem, that gap is the whole game.

The obvious assumption is that the leaders had a better model, or a feature I'd missed. So instead of guessing, I ran the experiment to its end: I tested every distinct model family I could build, measured each one against the others, and watched what the ensemble actually did with them. The result was not a missing trick. It was a **frontier** — a hard, monotonic tradeoff that no model on this data escapes.

This writeup is the map of that frontier: the levers that worked, the ones that didn't, two neural networks built from scratch, a tabular foundation model, and a closing finding sharp enough to explain the gap completely.

9

MODEL FAMILIES
TESTED

+13_{bps}

BIGGEST SINGLE
LEVER

0.92

LOWEST
CORRELATION
REACHED

4×

STACKING TIES
BLENDING

02 - THE CLIMB

From the 70th percentile to the 15th

The journey up the leaderboard was driven by a handful of decisive moves, not by incremental tuning. Each row below is a real submission.

SUBMISSION	OOF	LB	PERCENTILE
LightGBM baseline	0.94388	0.94432	~70%
9-model GBDT blend	0.95061	0.94993	~32%
3-model + ORIG-concat	0.95185	0.95119	~24%
RealMLP (from scratch)	0.95356	0.95338	~18%
RealMLP + XGB blend	~0.95470	0.95390	~15%

The biggest lever wasn't a model — it was data

The single largest jump (+13 bps) came from **ORIG-concat**: the competition data was synthesized from a real F1 strategy dataset, so I concatenated those ~101k original rows into each training fold while keeping validation on pure competition data. One ORIG-concat model beat my entire nine-model GBDT blend. *Data strategy outscores model count.*

LESSON

Before reaching for a bigger ensemble, find the lever in the data. A single well-fed model repeatedly beat a committee of starved ones.

Cross-family diversity is the real axis

The second jump came from blending a neural network with the trees. A RealMLP — built from scratch, not imported — alone beat the GBDT blend, and a rank-normalized **RealMLP + XGBoost** blend became the validated best. Tree splits and dense neural mixing make *different* errors; averaging them cancels noise. This is the moment the diversity question first appeared.

03 - BUILDING THE ENGINE

RealMLP, from components

Rather than `pip install` a tabular neural net, I reconstructed a RealMLP from its building blocks — the better to understand *why* it works. The architecture's distinctive idea is the **PBLD embedding** (Periodic Basis with Learned Decay): each numeric feature is projected through cosine waves at learned frequencies, letting the network represent periodic and threshold-like structure that a plain linear layer cannot.

```
# each numeric feature -> cosine waves at learned frequencies,
# then a small learned transform. captures thresholds & periodicity.
class PBLDEmbedding(nn.Module):
    def forward(self, x):
        per = torch.cos(2*math.pi*(x.unsqueeze(-1)*self.w1 + self.b1))
        tr = self.act(torch.einsum("bkfh,kfhd->bkfd", per, self.w2) + self.b2)
        return torch.cat([x.unsqueeze(-1), tr], dim=-1).flatten(start_dim=2)
```

Other components mattered too, and each taught something:

- ◆ **A 16-member internal ensemble** averaged at inference, batched efficiently with `einsum` rather than a Python loop.
- ◆ **Scheduled everything** — learning rate (flat-then-cosine), dropout (`expm4t` decay), and label smoothing all follow schedules that must *complete* where the model peaks. Stretching them over too many epochs scored *worse* (0.95256 at 8 epochs vs 0.95356 at 5) because they peaked mid-training.
- ◆ **Multi-seed averaging** reliably bought +5 bps by cutting neural variance — three seeds, predictions averaged.

COUNTER-INTUITIVE FINDING

More training was worse. The five-epoch model beat the eight-epoch model because its schedules finished exactly where the model converged. Schedule length is a hyperparameter, not an afterthought.

Stacking ties blending. Every time.

With a pool of strong models, the natural next move is a meta-model — stacking — to learn the optimal combination. I built it properly: logit-space meta-features (turning bounded probabilities into unbounded log-odds, plus per-row mean/std/min/max/range), then a cross-validated logistic-regression meta-learner sweeping regularization strength.

```
// LOGIT-SPACE META-FEATURES FOR THE STACKER
```

```
def logit(p): return np.log(p/(1-p))    # unbounded log-odds
L = logit(np.clip(P, 1e-6, 1-1e-6))
feats = np.hstack([L, L.mean(1,keepdims=True), L.std(1,keepdims=True),
                  L.min(1,keepdims=True), L.max(1,keepdims=True)])
```

The result, found **four separate times** across different model pools: stacking ties a simple rank-blend, to within 0.1 bp. In the five-model run, the regularization sweep landed every setting from $C=0.01$ to $C=3.0$ at the identical 0.95436–0.95437.

When one model dominates and the bases are correlated, a smarter combiner has nothing to exploit. The meta-model just rediscovers the blend.

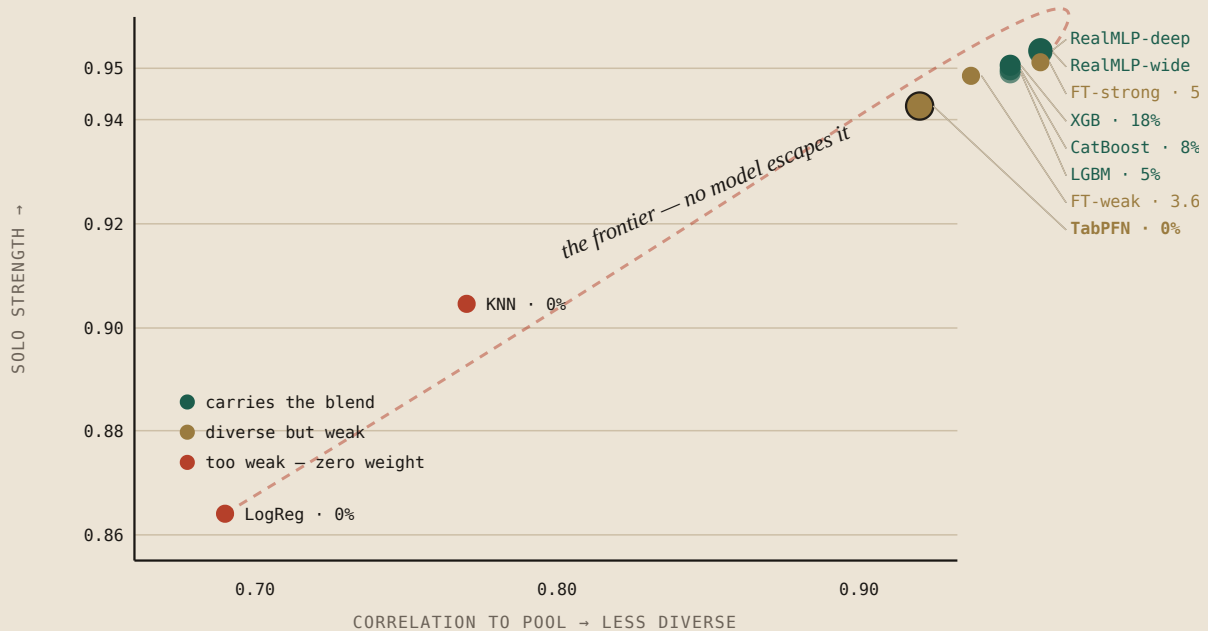
This is the first appearance of the wall. The combiner can't add what the models don't independently contain. Which raises the real question — *how independent are they?*

The diversity–strength frontier

To break a correlated ensemble, you need a model that is both **strong** (high solo accuracy) and **diverse** (low correlation to the pool). I tested nine families looking for that combination. Here is every one, plotted.

Every model lands on a downward curve

x: rank-correlation to the pool (lower = more diverse) · y: solo OOF AUC (higher = stronger)



The pattern is monotonic and admits no exception: **as a model becomes more diverse, it becomes weaker.** The strong models (RealMPLP, XGB) are tightly correlated at ~0.96 and carry the blend. Everything genuinely decorrelated — KNN, LogReg, and the foundation model — is too weak to earn any weight. There is no point in the strong-and-diverse corner, because the signal in this data is *unimodal*: every capable model converges on the same predictive structure.

MODEL	SOLO OOF	MIN CORR	BLEND WT	VERDICT
RealMLP-deep	0.95339	0.96	35%	carries
RealMLP-wide	0.95335	0.96	34%	carries
XGBoost	0.95071	0.95	18%	carries
CatBoost	0.94984	0.95	8%	carries
LightGBM	0.94981	0.95	5%	carries
FT-Transformer (strong)	0.95108	0.96	5.6%	strong→correlated
FT-Transformer (weak)	0.94842	0.937	3.6%	diverse, weak
TabPFN-2.5	0.94272	0.92	0%	most diverse, too weak
KNN	0.90462	0.77	0%	far too weak
Logistic Reg.	0.86414	0.69	0%	weakest

06 — THE KEYSTONE

A foundation model stress-tests the wall

TabPFN-2.5 was the single best theoretical shot at breaking the frontier. It's a transformer **pre-trained on millions of synthetic tabular datasets** — it does classification by in-context learning, with an inductive bias shaped by everything it

saw in pre-training rather than by what it can extract from this one dataset. If anything could sit in the strong-and-diverse corner, it was this.

It delivered exactly the diversity its pre-training promised:

0.92

CORR TO REALMLP — A
RECORD LOW

0.943

SOLO OOF — TOO WEAK

0%

BLEND WEIGHT ASSIGNED

The most diverse model in the entire investigation — and it earned **zero weight**. At 0.943 solo, roughly 11 bps below the RealMLPs, the blend optimizer decided that even genuinely orthogonal information wasn't worth a single percent when it arrived in a model that weak. (A practical note: at full scale, test inference on 188k rows projected to 43 hours — TabPFN's $O(\text{context} \times \text{query})$ cost forced a subsampling compromise that further capped its strength. Part of the weakness is the model; part is the data scale.)

WHY THIS IS THE KEYSTONE

The strongest tool available for producing diversity produced record diversity — and the frontier held anyway. That's not a null result. It's the cleanest possible confirmation that the tradeoff is structural, not a failure of effort.

07 — WHAT DIDN'T WORK

The negative results are the point

A rigorous investigation reports its failures with the same precision as its wins. Each of these was a genuine hypothesis, tested with honest cross-validation and leak-free

discipline.

Direct Driver / Race target encoding — -7.8 bps

A high-scoring public notebook leaned heavily on target-encoding the high-cardinality Driver and Race columns. Rebuilt *leak-free* (encoding fit inside each fold only), it made XGBoost slightly *worse* — the signal was redundant with what ORIG-concat and the combo encodings already captured. The public notebook's score came from a target leak that produced a fake 1.0 OOF AUC, not from the technique.

The "neighbor pit" feature — not reconstructable

A peer-reviewed paper on real F1 telemetry showed that a driver's pit probability roughly doubles when the car directly ahead pits — the classic undercut/overcut dynamic. A genuinely promising domain feature. But a feasibility check revealed the synthetic data destroyed the structure needed to build it: 96% of race-lap snapshots had duplicate track positions (61 cars sharing ~18 distinct positions), so "the car directly ahead" was ambiguous. Forced in anyway, it cost -14 bps — noise, exactly as predicted.

DISCIPLINE

Every tempting lever above came from a source that *looked* authoritative — a top public notebook, an academic paper. Testing them honestly, rather than trusting the source, is what kept the solution validated.

So what was the difference?

If no model breaks the frontier, where do the top scores come from? I dissected four high-scoring public solutions to find out.

- ◆ **A 0.9544 LightGBM** — a target leak. Leave-one-out encoding computed on the full train set before CV produced a fake OOF AUC of 1.00000.
- ◆ **A 0.95452 "blend of blends"** — arithmetic combinations of 27 finished submission files, 21 of them downloaded from other competitors. Pure public-leaderboard overfitting via "rank-remap" and "micro-intervention" tricks that did nothing measurable.
- ◆ **A 0.9544 two-layer stack** — stacked other people's published OOF files. The elaborate stacking did nothing over a simple average; the lift came entirely from a larger pool of *others'* models.
- ◆ **A 0.95454 submission** — its own model was an XGBoost *regressor* on a ranking metric with 935 one-hot features and feature code referencing columns that don't exist in this dataset. The score came from a downloaded submission blended at weight 2.9 against the author's own model at weight 0.000001.

The gap between an honest 0.95390 and the public 0.9545s is not a skill gap. It's the difference between a self-built, validated solution and a pool of borrowed public outputs.

This matters beyond etiquette. Pooled, public-LB-tuned solutions frequently **compress on the private leaderboard** when the hidden test split rewards genuine generalization over noise-fitting. A clean, well-validated 0.95390 is the kind of solution that *holds*.

What carries to the next competition

- 1 **Find the data lever first.** ORIG-concat (+13 bps) beat every modeling decision. A single well-fed model outscored a nine-model committee.
- 2 **Diversity is cross-family, and it's strong-vs-weak.** Neural-vs-tree is the real axis; a second model of the same family adds almost nothing (two RealMLPs correlated 0.96; FT-Transformer and trees, 0.95).
- 3 **Stacking only beats blending when the bases are genuinely decorrelated.** When one model dominates a correlated pool, the meta-model just rediscovers the average. Confirmed four times.
- 4 **Schedules must complete where the model peaks.** Five epochs beat eight because the LR/dropout/label-smoothing schedules finished on time.
- 5 **Architecture variation beats seed variation** for cheap neural diversity (wide vs deep RealMLP correlated 0.96; same-arch seeds, 0.99+).
- 6 **Trust validation over sources.** Every leaked or borrowed "lever" came dressed as authority. Honest in-fold testing is the only defense.
- 7 **The diversity–strength tradeoff is real and may be intrinsic to a dataset.** Sometimes the ceiling is the data's, not yours — and proving that is itself a result.

The ceiling was real. Mapping it was the work.

Final validated submissions: a RealMLP + XGBoost rank-blend at **0.95390**, with a multi-seed RealMLP at 0.95357 as a robust fallback. A top-15% finish — every component built, understood, and validated from scratch.

Nine model families · two from-scratch neural architectures · one foundation model · zero borrowed submissions